

Time series data

Page No.:

Date: / /

Index	Temp (θ)
D1	25
D2	13
D3	12
D4	31
D5	43

(Exponentially
Weighted
Moving Average)

to find trends in
time series data by
giving exponentially
decreasing weight to
older observations

$$V_t = \beta V_{t-1} + (1-\beta)\theta_t$$

$\beta \rightarrow 0 < \beta < 1$

β how much weight to
give to the past data

($\beta = 0.9$) preferred

✓ Directly

$$V_0 = \theta_0$$

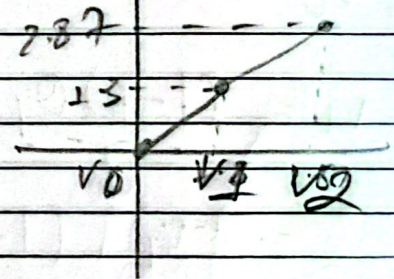
$$V_1 = 0.9 \times V_0 + 0.1 \times 13$$

$$= 0 + 1.3$$

$$= 1.3$$

$$V_2 = 0.9 \times 1.3 + 0.1 \times 17$$

$$= 2.87$$



upto

V_5

Page No.:

Date: / /

Mathematical Induction,

Page No.:

Date: / /

$$V_t = \beta V_{t-1} + (1-\beta) \theta_t$$

Now,

$$V_0 = 0 \quad \leftarrow \text{let}$$

$$(i) \quad V_1 = \underbrace{(1-\beta)}_1 \theta_1$$

$$(ii) \quad V_2 = \beta V_1 + (1-\beta) \theta_2$$

$$(iii) \quad \underbrace{V_2}_{\downarrow} = \beta(1-\beta)\theta_1 + (1-\beta)\theta_2$$

$$(iv) \quad V_3 = \beta V_2 + (1-\beta)\theta_3 \\ = \beta^2(1-\beta)\theta_1 + \beta(1-\beta)\theta_2 + (1-\beta)\theta_3$$

$$(v) \quad V_4 = \beta V_3 + (1-\beta)\theta_4 \\ = \beta^3(1-\beta)\theta_1 + \beta^2(1-\beta)\theta_2 + \beta(1-\beta)\theta_3 \\ + (1-\beta)\theta_4$$

$$= (1-\beta) [\beta^3\theta_1 + \beta^2\theta_2 + \beta\theta_3 + \theta_4]$$

Since $\beta (0 < \beta < 1)$

So, less weight to past data,
& more weight to
present data

Page No.:

Date: / /